

LAB # 6

AGGREGATION FUNCTIONS

Objective:

To learn how to use SQL aggregate functions (COUNT, AVG, MAX, MIN and SUM) to summarize data.

Introduction:

Aggregate functions perform a calculation on a set of rows and return a single summarized value. They are used to answer questions about a whole dataset, such as how many records exist, what the average or total of a column is, and what the highest and lowest values are. The functions used in this lab are:

- **COUNT()** : returns the number of rows.
- **AVG()** : returns the average value of a numeric column.
- **MAX() and MIN()** : return the highest and lowest values of a column.
- **SUM()** : returns the total of a numeric column.

Aggregate functions can also be combined with the GROUP BY clause to get a separate summary for each group of rows, such as the number of students in each department.

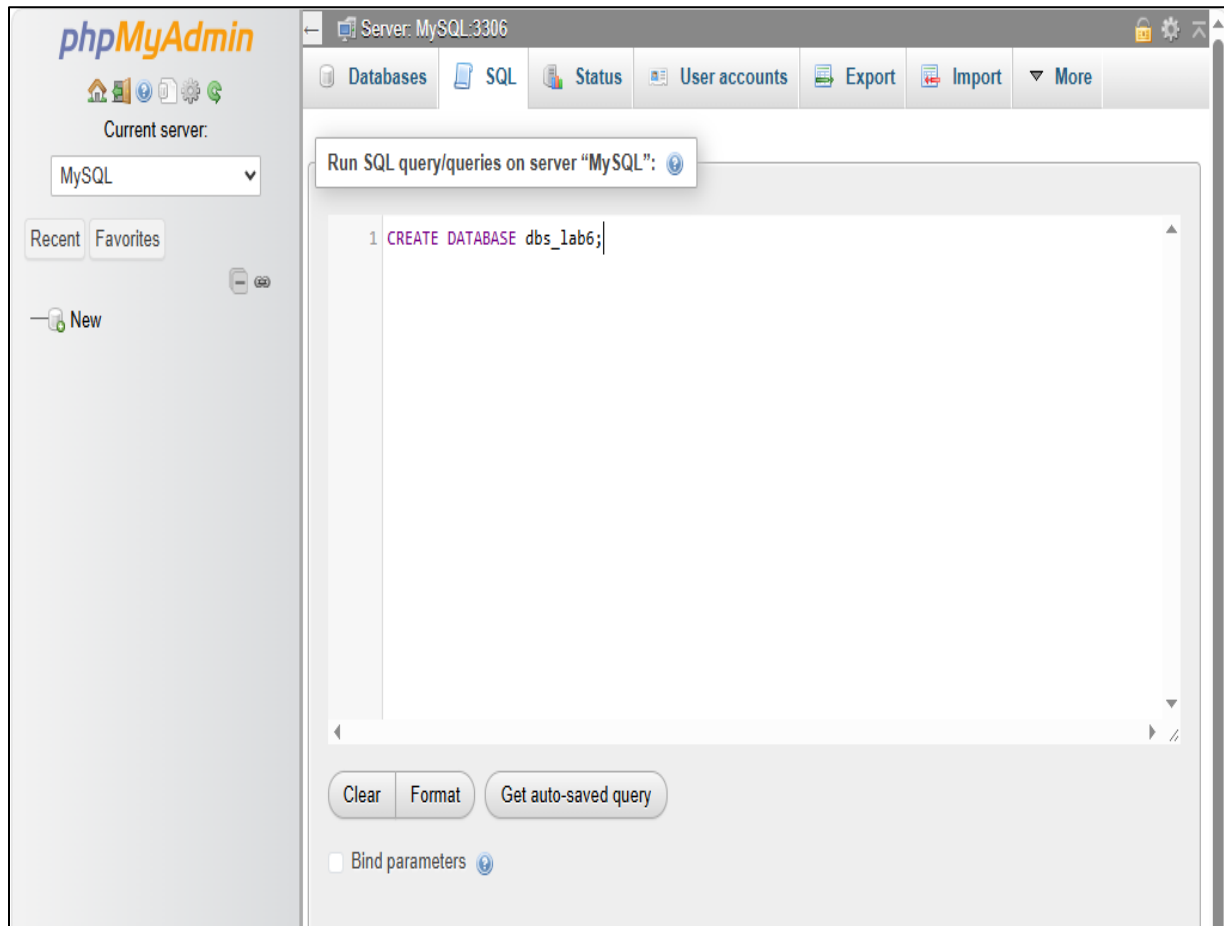
Procedure:

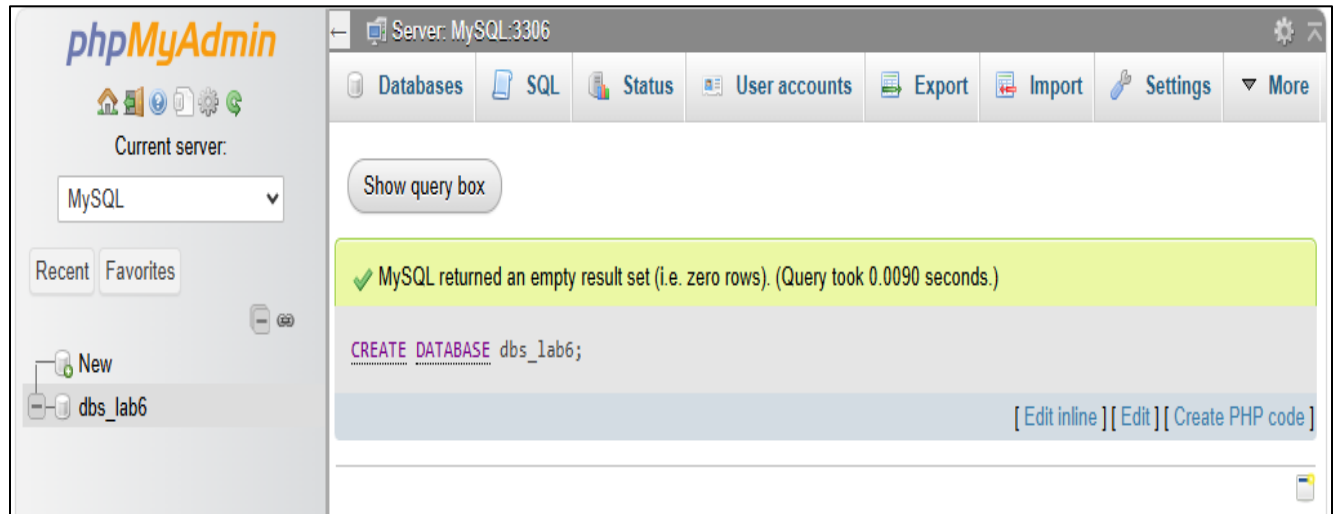
1. Start WampServer and open phpMyAdmin in the browser (localhost/phpmyadmin).
2. Create the database dbs_lab6 and select it.
3. Create the students table and insert eight records.
4. Create the fees table with a foreign key on student_id and insert eight records.
5. Run the COUNT, AVG, MAX/MIN, SUM and COUNT with GROUP BY queries one by one in the SQL tab.
6. Take a screenshot of each query and of its output.

LAB TASK**Step 1: Create Database**

A database named "dbs_lab6" was created and then selected for use.

```
CREATE DATABASE dbs_lab6;
```



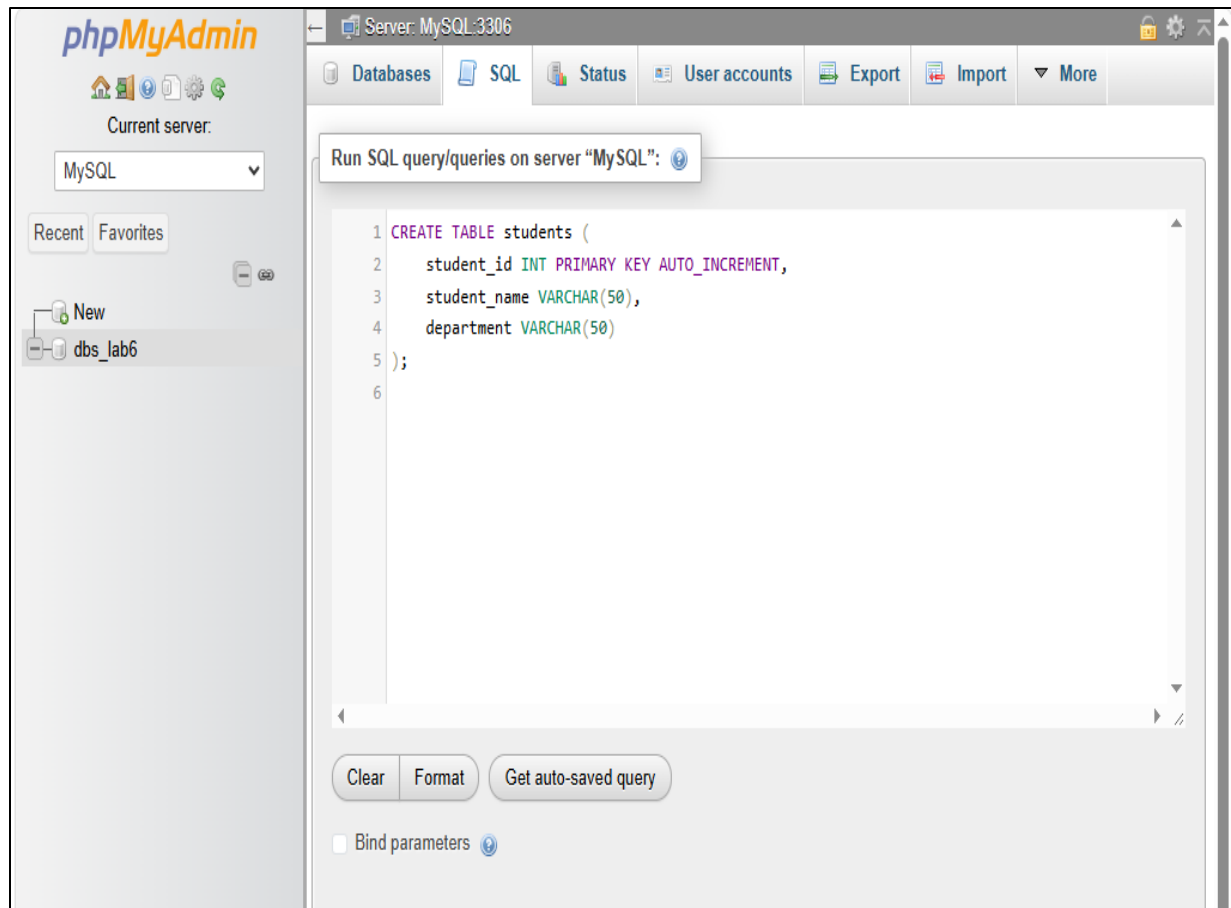


Output: Database created successfully

Step 2: Create the students Table

The students table was created with student_id as the primary key.

```
CREATE TABLE students (  
    student_id INT PRIMARY KEY AUTO_INCREMENT,  
    student_name VARCHAR(50),  
    department VARCHAR(50)  
);
```



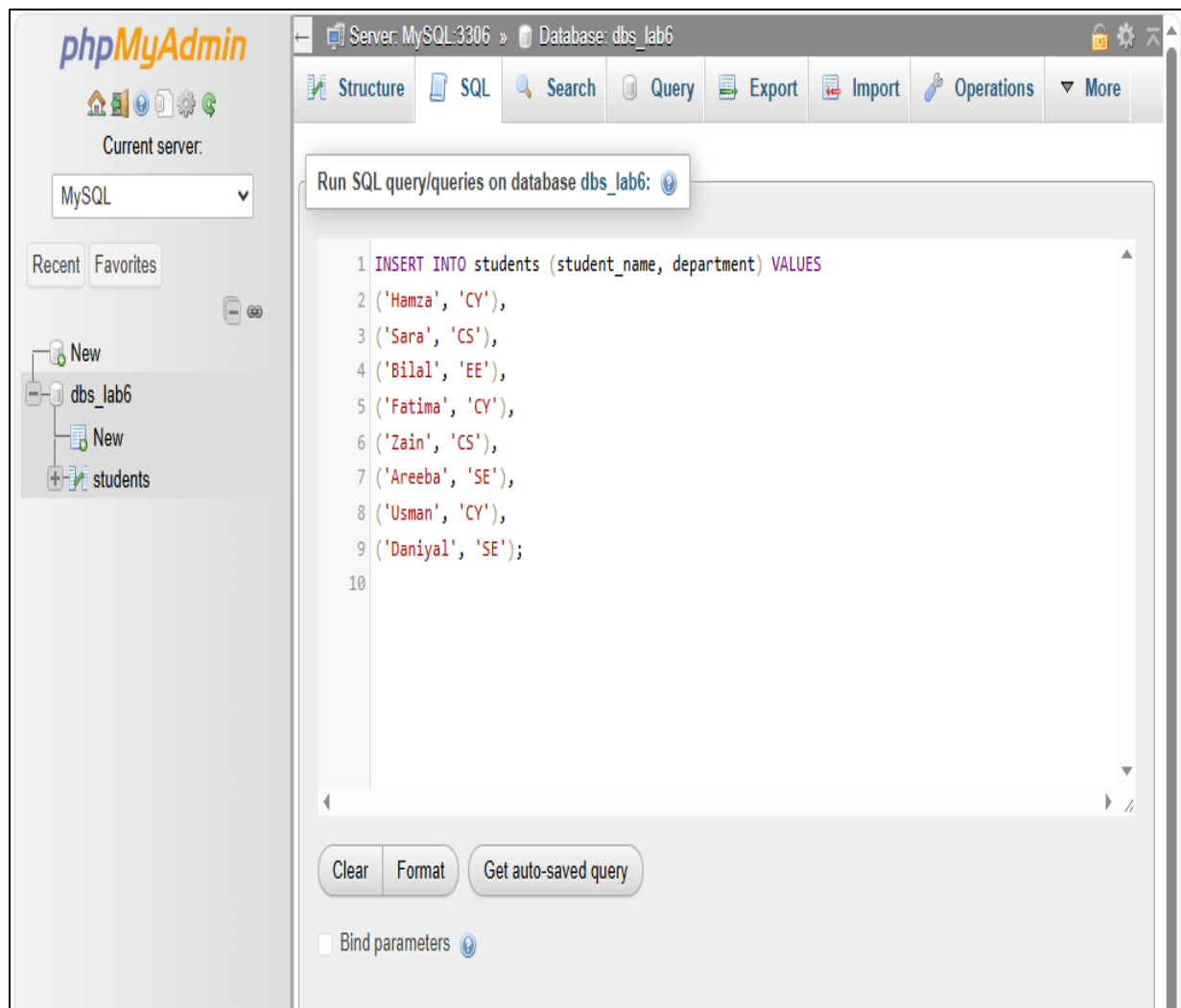
The screenshot displays the phpMyAdmin web interface. The top navigation bar shows the server as 'MySQL:3306' and the database as 'dbs_lab6'. The left sidebar contains a tree view of the database structure, with 'dbs_lab6' expanded to show a 'New' button and a 'students' table. The main content area features a toolbar with options like 'Structure', 'SQL', 'Search', 'Query', 'Export', 'Import', 'Operations', and 'More'. Below the toolbar, a 'Show query box' button is visible. The central area shows a green success message: 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0136 seconds.)'. Below this, the SQL query used to create the table is displayed: `CREATE TABLE students ( student_id INT PRIMARY KEY AUTO_INCREMENT, student_name VARCHAR(50), department VARCHAR(50));`. At the bottom of the query box, there are links for '[Edit inline]', '[Edit]', and '[Create PHP code]'.

Output: students table created successfully

Step 3: Insert Records into students

Eight student records were inserted into the students table.

```
INSERT INTO students (student_name, department) VALUES
('Hamza', 'CY'),
('Sara', 'CS'),
('Bilal', 'EE'),
('Fatima', 'CY'),
('Zain', 'CS'),
('Areeba', 'SE'),
('Usman', 'CY'),
('Daniyal', 'SE');
```



The screenshot shows the phpMyAdmin interface. On the left, the 'Current server' is set to 'MySQL'. The database 'dbs_lab6' is selected, and the 'students' table is visible in the left sidebar. The main panel displays the results of a query: '8 rows inserted. Inserted row id: 8 (Query took 0.0008 seconds.)'. Below this, the SQL query is shown: `INSERT INTO students (student_name, department) VALUES ('Hamza', 'CY'), ('Sara', 'CS'), ('Bilal', 'EE'), ('Fatima', 'CY'), ('Zain', 'CS'), ('Areeba', 'SE'), ('Usman', 'CY'), ('Daniyal', 'SE');`. Links for '[Edit inline]', '[Edit]', and '[Create PHP code]' are provided.

The screenshot shows the phpMyAdmin interface with the 'students' table selected. The main panel displays the results of a query: 'Showing rows 0 - 7 (8 total, Query took 0.0010 seconds.)'. Below this, the SQL query is shown: `SELECT * FROM `students``. The table data is displayed in a grid format with columns: student_id, student_name, and department. The data is as follows:

student_id	student_name	department
1	Hamza	CY
2	Sara	CS
3	Bilal	EE
4	Fatima	CY
5	Zain	CS
6	Areeba	SE
7	Usman	CY
8	Daniyal	SE

At the bottom, there are links for '[Edit inline]', '[Edit]', '[Explain SQL]', '[Create PHP code]', and '[Refresh]'. There are also checkboxes for 'Show all', 'Number of rows: 25', 'Filter rows: Search this table', and 'Sort by key: None'. At the bottom, there are links for 'Check all', 'With selected: Edit', 'Copy', 'Delete', and 'Export'.

Output: Eight records inserted (verified with `SELECT * FROM students;`)

Step 4: Create the fees Table

The fees table was created and linked to the students table through a foreign key.

```
CREATE TABLE fees (  
    fee_id INT PRIMARY KEY AUTO_INCREMENT,  
    student_id INT,  
    marks INT,  
    FOREIGN KEY (student_id) REFERENCES students(student_id)  
);
```

The screenshot shows the phpMyAdmin web interface. The top navigation bar indicates the current context: Server: MySQL:3306, Database: dbs_lab6, and Table: students. The left sidebar shows the database structure with 'dbs_lab6' selected and 'students' as the current table. The main panel displays the 'SQL' tab with a text area containing the SQL query to create the 'fees' table. The query is as follows:

```
1 CREATE TABLE fees (  
2     fee_id INT PRIMARY KEY AUTO_INCREMENT,  
3     student_id INT,  
4     marks INT,  
5     FOREIGN KEY (student_id) REFERENCES  
6     students(student_id)  
7 );
```

Below the text area, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and '<<'. There is also a 'Get auto-saved query' button and a 'Bind parameters' checkbox.

The screenshot displays the phpMyAdmin web interface. The top navigation bar shows the current context: Server: MySQL:3306, Database: dbs_lab6, and Table: students. Below this, a menu bar includes options like Browse, Structure, SQL, Search, Insert, Export, Import, and Privileges. The left sidebar shows the database structure, with 'dbs_lab6' expanded to show tables 'fees' and 'students'. The main content area shows a successful query execution result: 'MySQL returned an empty result set (i.e. zero rows). (Query took 0.0125 seconds.)'. Below this, the SQL query used to create the 'fees' table is displayed: `CREATE TABLE fees ( fee_id INT PRIMARY KEY AUTO_INCREMENT, student_id INT, marks INT, FOREIGN KEY (student_id) REFERENCES students(student_id) );`. At the bottom of the query area, there are links for [Edit inline], [Edit], and [Create PHP code].

Output: fees table created successfully

Step 5: Insert Records into fees

Eight records were inserted into the fees table.

```
INSERT INTO fees (student_id, marks) VALUES  
(1, 78), (2, 91), (3, 66), (4, 84),  
(5, 88), (6, 72), (7, 95), (8, 80);
```

The screenshot shows the phpMyAdmin interface for a MySQL database named 'dbs_lab6'. The 'Table: fees' is selected. The SQL query editor contains the following query:

```
1 INSERT INTO fees (student_id, marks) VALUES  
2 (1, 78), (2, 91), (3, 66), (4, 84),  
3 (5, 88), (6, 72), (7, 95), (8, 80);  
4
```

The query is executed, and the results are displayed in a table with the following columns: fee_id, student_id, and marks. The results show 8 rows of data.

fee_id	student_id	marks
1	1	78
2	2	91
3	3	66
4	4	84
5	5	88
6	6	72
7	7	95
8	8	80

The interface also includes a sidebar with the database structure, a top navigation bar with options like Browse, Structure, SQL, Search, Insert, Export, Import, and More, and a bottom section with buttons for SELECT *, SELECT, INSERT, UPDATE, DELETE, Clear, Format, and a checkbox for Bind parameters.

The screenshot shows the phpMyAdmin interface. On the left, the server is set to MySQL, and the database 'dbs_lab6' is selected. The table 'fees' is highlighted. The main panel shows a green message: '8 rows inserted. Inserted row id: 8 (Query took 0.0009 seconds.)'. Below this, the SQL query is displayed: `INSERT INTO fees (student_id, marks) VALUES (1, 78), (2, 91), (3, 66), (4, 84), (5, 88), (6, 72), (7, 95), (8, 80);`. The top navigation bar includes links for Browse, Structure, SQL, Search, Insert, Export, Import, and Privileges.

The screenshot shows the phpMyAdmin interface with the 'fees' table selected. A green message indicates: 'Showing rows 0 - 7 (8 total, Query took 0.0010 seconds.)'. The SQL query `SELECT * FROM `fees`` is shown. Below the query, there are controls for 'Show all', 'Number of rows' (set to 25), 'Filter rows' (Search this table), and 'Sort by key' (None). An 'Extra options' button is also present. The table data is displayed with columns 'fee_id', 'student_id', and 'marks'. Each row includes checkboxes for 'Edit', 'Copy', and 'Delete'.

	fee_id	student_id	marks
☐ Edit ☐ Copy ☐ Delete	1	1	78
☐ Edit ☐ Copy ☐ Delete	2	2	91
☐ Edit ☐ Copy ☐ Delete	3	3	66
☐ Edit ☐ Copy ☐ Delete	4	4	84
☐ Edit ☐ Copy ☐ Delete	5	5	88
☐ Edit ☐ Copy ☐ Delete	6	6	72
☐ Edit ☐ Copy ☐ Delete	7	7	95
☐ Edit ☐ Copy ☐ Delete	8	8	80

At the bottom, there are controls for 'Check all', 'With selected', and actions: Edit, Copy, Delete, and Export.

Output: Eight records inserted (verified with `SELECT * FROM fees;`)

Step 6: COUNT

The COUNT function was used to find the total number of students.

```
SELECT COUNT(*) AS total_students  
FROM students;
```

The screenshot displays the phpMyAdmin web interface. The top navigation bar shows the current context: Server: MySQL 3306, Database: dbs_lab6, and Table: students. Below this, a toolbar contains buttons for Browse, Structure, SQL, Search, Insert, Export, Import, and More. The left sidebar shows the database structure with 'dbs_lab6' selected, containing tables 'fees' and 'students'. The main area is titled 'Run SQL query/queries on table dbs_lab6.students:'. It features a text input field with the SQL query:
1 SELECT COUNT(*) AS total_students
2 FROM students;
3
To the right of the query field is a dropdown menu showing the table structure: student_id, student_name, and department. Below the query field, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and '<<'. A 'Get auto-saved query' button is also present. At the bottom, there is a checkbox for 'Bind parameters'.

The screenshot shows the phpMyAdmin web interface. The top navigation bar indicates the current context: Server: MySQL:3306, Database: dbs_lab6, and Table: students. The left sidebar shows the database structure with 'dbs_lab6' selected, containing tables 'fees' and 'students'. The main panel displays a message: 'Your SQL query has been executed successfully. (Query took 0.0002 seconds.)' followed by the SQL query: `SELECT COUNT(*) AS total_students FROM students;`. Below the query, the result is shown as a table with one row:

total_students
8

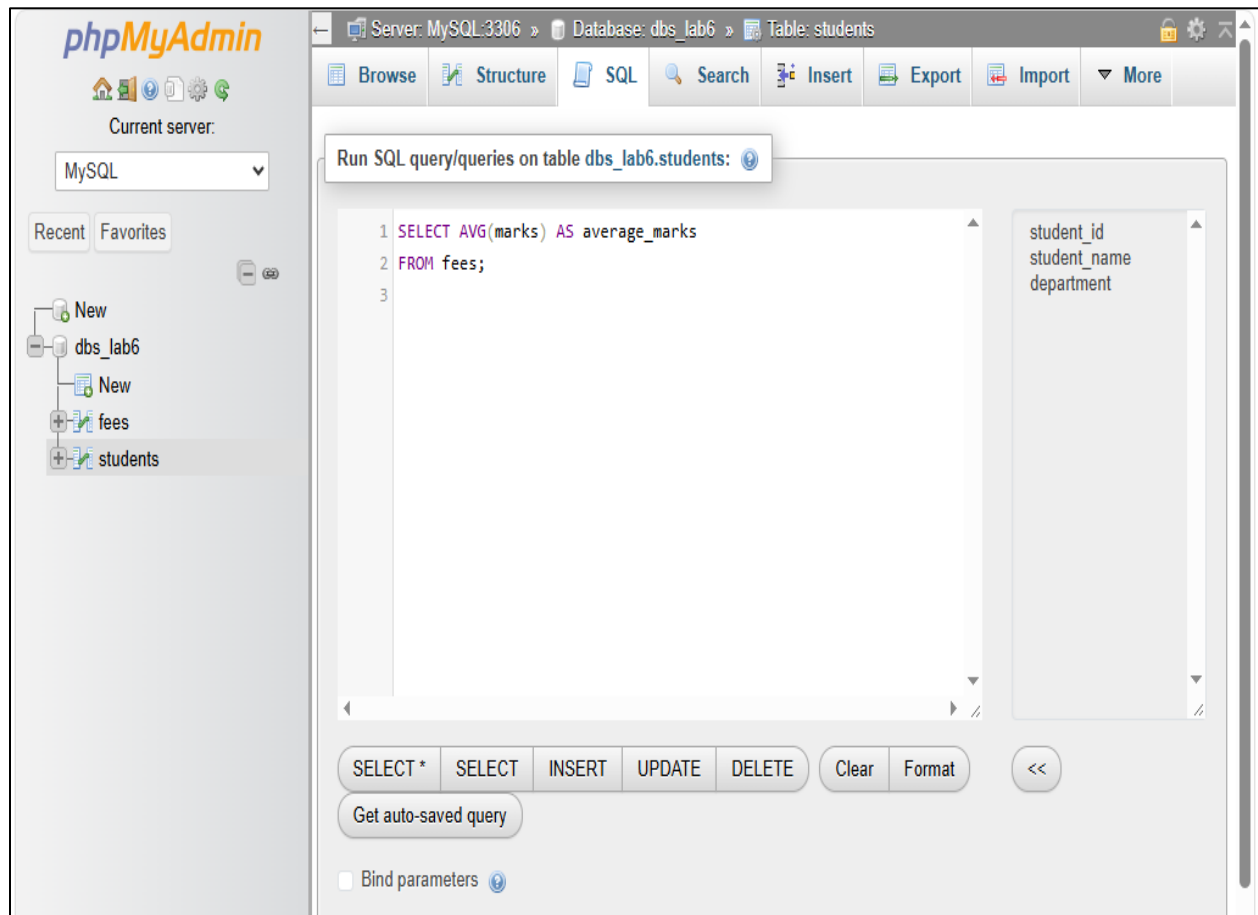
. The bottom of the main panel includes a 'Query results operations' bar with buttons for Print, Copy to clipboard, Export, Display chart, and Create view.

Output: Total number of students

Step 7: AVG

The AVG function was used to find the average marks.

```
SELECT AVG(marks) AS average_marks  
FROM fees;
```



The screenshot displays the phpMyAdmin web interface. The top navigation bar shows the server as 'MySQL:3306', the database as 'dbs_lab6', and the selected table as 'students'. The main menu includes options like 'Browse', 'Structure', 'SQL', 'Search', 'Insert', 'Export', 'Import', and 'More'. The left sidebar shows the database structure with 'dbs_lab6' containing tables 'fees' and 'students'. The central area is titled 'Run SQL query/queries on table dbs_lab6.students:' and contains a text input field with the following SQL query:

```
1 SELECT AVG(marks) AS average_marks  
2 FROM fees;  
3
```

Below the query input, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and '<<'. A 'Get auto-saved query' button is also present. At the bottom, there is a checkbox for 'Bind parameters'.

The screenshot shows the phpMyAdmin web interface. On the left is a sidebar with the 'phpMyAdmin' logo and a tree view of the database structure. The tree shows a database named 'dbs_lab6' containing two tables: 'fees' and 'students'. The 'fees' table is selected. The main panel at the top has tabs for 'Browse', 'Structure', 'SQL', 'Search', 'Insert', 'Export', 'Import', and 'Privileges'. The 'SQL' tab is active, displaying a 'Show query box' button and a message: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' Below this, a green status bar indicates 'Showing rows 0 - 0 (1 total, Query took 0.0080 seconds.)'. The SQL query entered is: `SELECT AVG(marks) AS average_marks FROM fees;`. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A control bar shows 'Show all', 'Number of rows: 25', and a 'Filter rows' search box. An 'Extra options' button is also present. The query result is displayed in a table with one row:

average_marks
81.7500

. Below the table, there is another control bar with 'Show all', 'Number of rows: 25', and a 'Filter rows' search box. At the bottom, a 'Query results operations' bar contains links for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

Output: Average marks

Step 8: MAX and MIN

The MAX and MIN functions were used to find the highest and lowest marks.

```
SELECT MAX(marks) AS highest_marks,  
       MIN(marks) AS lowest_marks  
FROM fees;
```

The screenshot displays the phpMyAdmin web interface. The top navigation bar shows the server as 'MySQL:3306', the database as 'dbs_lab6', and the selected table as 'students'. Below this, a toolbar contains buttons for 'Browse', 'Structure', 'SQL', 'Search', 'Insert', 'Export', 'Import', and 'More'. The left sidebar shows the 'Current server' as 'MySQL' and a tree view of the database structure with 'dbs_lab6' expanded, showing tables 'fees' and 'students'. The main area is titled 'Run SQL query/queries on table dbs_lab6.students:' and contains a text editor with the following SQL query:

```
1 SELECT MAX(marks) AS highest_marks,  
2     MIN(marks) AS lowest_marks  
3 FROM fees;  
4
```

To the right of the query editor is a list of table columns: 'student_id', 'student_name', and 'department'. At the bottom of the interface, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and '<<'. Below these buttons is a 'Get auto-saved query' button and a checkbox labeled 'Bind parameters'.

The screenshot shows the phpMyAdmin interface. On the left is a sidebar with a tree view containing 'New', 'db6_lab6', 'New', 'fees', and 'students'. The 'fees' table is selected. The main area has a top navigation bar with 'Browse', 'Structure', 'SQL', 'Search', 'Insert', 'Export', 'Import', and 'Privileges'. Below this is a 'Show query box' button. A warning message states: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' Below the warning, a green bar indicates 'Showing rows 0 - 0 (1 total, Query took 0.0074 seconds.)'. The SQL query is: `SELECT MAX(marks) AS highest_marks, MIN(marks) AS lowest_marks FROM fees;`. Below the query are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A control bar shows 'Show all', 'Number of rows: 25', and a 'Filter rows' search box. An 'Extra options' button is also present. The query results are displayed in a table with two columns: 'highest_marks' and 'lowest_marks'. The values are 95 and 66 respectively. Below the table is another control bar with 'Show all', 'Number of rows: 25', and a 'Filter rows' search box. At the bottom, a 'Query results operations' bar contains links for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

Server: MySQL:3306 » Database: db6_lab6 » Table: fees

Current server: MySQL

Recent Favorites

New
db6_lab6
New
fees
students

Structure SQL Search Insert Export Import Privileges

Show query box

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

✓ Showing rows 0 - 0 (1 total, Query took 0.0074 seconds.)

```
SELECT MAX(marks) AS highest_marks, MIN(marks) AS lowest_marks FROM fees;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

highest_marks	lowest_marks
95	66

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print Copy to clipboard Export Display chart Create view

Output: Highest and lowest marks

Step 9: SUM

The SUM function was used to find the total of all marks.

```
SELECT SUM(marks) AS total_marks  
FROM fees;
```

The screenshot displays the phpMyAdmin web interface. The top navigation bar shows the current server as 'MySQL' and the selected database as 'dbs_lab6'. The 'Table: fees' is selected. The left sidebar shows the database structure with 'dbs_lab6' containing tables 'fees' and 'students'. The main area is titled 'Run SQL query/queries on table dbs_lab6.fees:'. The SQL query editor contains the following code:

```
1 SELECT SUM(marks) AS total_marks  
2 FROM fees;  
3
```

Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and '<<'. A 'Get auto-saved query' button is also present. At the bottom, there is a checkbox for 'Bind parameters' which is currently unchecked. On the right side of the query editor, a preview of the table structure is visible, showing columns: 'fee_id', 'student_id', and 'marks'.

The screenshot shows the phpMyAdmin web interface. On the left is a sidebar with the 'phpMyAdmin' logo and a tree view of the database structure. The tree shows a database named 'dbs_lab6' containing two tables: 'fees' and 'students'. The 'fees' table is currently selected. The main panel at the top has tabs for 'Browse', 'Structure', 'SQL', 'Search', 'Insert', 'Export', 'Import', and 'Privileges'. The 'SQL' tab is active, displaying a 'Show query box' button and a warning message: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' Below this, a green status bar indicates 'Showing rows 0 - 0 (1 total. Query took 0.0004 seconds.)'. The SQL query entered is: `SELECT SUM(marks) AS total_marks FROM fees;`. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A control bar shows 'Show all', 'Number of rows: 25', and a 'Filter rows' search box. The query result is displayed in a table with one row:

total_marks
654

. Below the table is another control bar with 'Show all', 'Number of rows: 25', and a 'Filter rows' search box. At the bottom, a 'Query results operations' section contains links for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

Output: Total marks

Step 10: COUNT with GROUP BY

COUNT was combined with GROUP BY to count the students in each department.

```
SELECT department, COUNT(*) AS total_students  
FROM students  
GROUP BY department;
```

The screenshot shows the phpMyAdmin web interface. The top navigation bar includes links for Browse, Structure, SQL, Search, Insert, Export, Import, and More. The current server is MySQL, and the selected database is db6. The left sidebar shows the database structure with tables fees and students. The main area displays the SQL query editor with the following query:

```
1 SELECT department, COUNT(*) AS total_students  
2 FROM students  
3 GROUP BY department;  
4
```

Below the query editor are buttons for SELECT *, SELECT, INSERT, UPDATE, DELETE, Clear, Format, and a double left arrow button. There is also a "Get auto-saved query" button and a "Bind parameters" checkbox.

The screenshot shows the phpMyAdmin interface. On the left is the sidebar with the database 'dbs_lab6' and tables 'fees' and 'students'. The main panel shows the 'Table: students' view. A message at the top states: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' Below this, a green status bar indicates 'Showing rows 0 - 3 (4 total, Query took 0.0012 seconds.)'. The SQL query entered is: `SELECT department, COUNT(*) AS total_students FROM students GROUP BY department;`. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A control bar shows 'Show all', 'Number of rows: 25', and a 'Filter rows' search box. The query results are displayed in a table with two columns: 'department' and 'total_students'. The results are: CY (3), CS (2), EE (1), and SE (2). Below the results, there is another control bar with 'Show all', 'Number of rows: 25', and a 'Filter rows' search box. At the bottom, there is a 'Query results operations' section with links for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

Server: MySQL:3306 » Database: dbs_lab6 » Table: students

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 3 (4 total, Query took 0.0012 seconds.)

```
SELECT department, COUNT(*) AS total_students FROM students GROUP BY department;
```

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Extra options

department	total_students
CY	3
CS	2
EE	1
SE	2

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

[Print](#) [Copy to clipboard](#) [Export](#) [Display chart](#) [Create view](#)

Output: Number of students in each department

Results:

Both tables were created and filled without any error, with eight records in each. The aggregate queries gave the following results:

- **COUNT:** the students table contains 8 students in total.
- **AVG:** the average marks of all students is 81.75.
- **MAX / MIN:** the highest mark is 95 and the lowest mark is 66.
- **SUM:** the total of all marks is 654.
- **GROUP BY:** CY has 3 students, CS has 2, SE has 2 and EE has 1, which adds up to the same 8 students counted earlier.

Conclusion:

This lab showed how aggregate functions summarize many rows into a single meaningful value, and how GROUP BY extends them to give a summary for each category. COUNT, AVG, MAX, MIN and SUM each answered a different question about the same data, and the department-wise counts matched the total student count, which confirms that the queries were consistent. These functions are the basis of reporting and analysis queries on larger datasets.